

Note Manager & Extractive Summarizer

Overview of Modules

Module	Component	Key Features
1	Vault & File Management	Directory ingestion, CRUD, tags, binary persistence
2	Inverted Index & Search	Custom index, phrase matching, ANSI highlighting
3	Extractive Summarizer	TF-IDF, LexRank graph centrality, summary export
4	Text Analytics & Insights	Metrics, top keywords, cosine similarity
5	Terminal REPL & Architecture	Command loop, exception handling

Module 1 — Vault and File Management

1.1 Automated Directory Ingestion

- Allows users to specify local disk folders/directories.
- Automatically scans and registers supported files:
 - Plain Text (.txt)
 - Markdown (.md)

1.2 Note Store — CRUD Operations

- Provides Create, Read, Update, and Delete operations through terminal commands.
- Supports soft deletion by moving deleted notes to a temporary Trash storage.
- Supports permanent deletion when explicitly requested.

1.3 Tag Manager

- Assigns custom tags and categories to notes.
- Supports multiple tags for individual notes.
- Tags can be used for filtering, searching, and summarization.

1.4 Optional Inline Tag Convention

- Allows tags to be specified directly within the first line of a note.
- The convention is optional and automatically registers the specified tags.

1.5 Binary Metadata Persistence

- Serializes metadata and index states into a custom binary format.
- Maintains persistent information required for efficient system operation.

Module 2 — High-Speed Inverted Index and Search Engine

2.1 Custom Inverted Index

- Implements a hand-coded inverted index from scratch.
- Maps unique keywords to their source files and line locations.
- Uses custom data structures such as linked lists, trees, hash-based structures, or dynamic arrays.

2.2 Advanced Querying and Phrase Matching

- Supports fast multi-file keyword searching.
- Provides exact multi-word phrase matching across the note collection.

2.3 Visual Highlighting and Filters

- Uses ANSI escape codes to highlight search matches in the terminal.
- Supports filtering search results based on custom tags and categories.

Module 3 — Advanced Extractive Summarizer

3.1 TF-IDF Vector Space Model

- Normalizes and tokenizes document text.
- Removes stop words.
- Calculates Term Frequency (TF) and Inverse Document Frequency (IDF).
- Generates TF-IDF representations across the note collection.

3.2 Graph-Based LexRank Summarization

- Represents sentences as nodes in a similarity graph.
- Determines edge weights using cosine similarity.
- Applies iterative power iteration to calculate sentence centrality.
- Selects highly central sentences to generate extractive summaries.

3.3 Flexible Summary Options

- Generates summaries for individual notes.
- Supports user-defined summary length constraints.
- Generates unified summaries from multiple notes or tagged categories.
- Exports generated summaries as new text/document files.

Module 4 — Text Analytics and Document Insights

4.1 Document Metrics

- Calculates:
 - Total word count
 - Sentence count
 - Average sentence length
 - Estimated reading time

4.2 Top Keyword Extraction

- Analyzes the note corpus to identify statistically significant keywords.
- Extracts and displays the top ten keywords.

4.3 Vector Similarity and Document Comparison

- Calculates pairwise cosine similarity between notes.
- Identifies highly similar or duplicate content.
- Detects overlapping topics across lecture notes and documents.

Module 5 — Terminal REPL and System Architecture

5.1 Interactive Command Loop

- Provides a continuous Read-Evaluate-Print Loop (REPL) interface.
- Supports structured terminal-based command execution for all major system operations.

5.2 Robust Exception and Error Handling

- Provides safe file I/O operations.
- Handles missing paths, invalid commands, and malformed input.
- Uses appropriate exception handling to prevent system crashes.

Overall System Architecture

